

ACADEMIC DETAILS				
Degree	Specialization	Institute	Year	CPI/%
B.Tech.	Computer Science and Engineering	IIT Gandhinagar	2023-Present	9.25/10
Class XII	Physics, Chemistry, Maths	B. K. D. Aldrich Public School	2021-2022	98.5%
Class X		B. K. D. Aldrich Public School	2019-2020	98%

PUBLICATIONS

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Why Multi-Source Audio Fails to Compose: A Geometric Diagnosis and Its Remedy
(Anurag Singh, Prajwal Singh, Prof. Shanmuganathan Raman)

[Accepted, BMVC 2026]

◦

Diagnosed compositional failure in multi-source A2I as **geometric displacement** of audio tokens in **CLIP space**, formalized through clustering analysis over intra-class compactness, angular alignment, and inter-token separation.

◦

Introduced a **mixed inference-time objective** combining **Attend-and-Excite** with an **A* spatial loss**, jointly enforcing token activation and spatial separation of cross-attention maps, resolving neglect and object fusion.

◦

Achieved upto **2.25×** higher **AIC** over **MACS** and **1.7×** lower **FID** over **AudioToken** with consistent **IIS** gains, ranking best on four of six metrics across **five** benchmarks spanning one to five audio sources.

INTERNSHIPS

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Data Science Intern – Process Intelligence
(Skan AI)

[May '26 – July '26]

◦

Improved the **signal to noise ratio** of captured event logs by filtering diagnosed noise, using a **rule based framework** to detect and filter **noisy events**, consistently reducing **process variants** and simplifying the resulting process graphs.

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Research Intern – Compositional Audio-to-Image Generation
(Prof. Shanmuganathan Raman, SRIP, IIT Gandhinagar) | [Reports](#)

[May '25 – July '25]

◦

Analyzed and benchmarked **20+** models including **AudioToken**, SonicDiffusion, Sound2Scene, and **MACS** on VGG-Sound, **IS3**, and curated datasets for audio-to-image (A2I) generation across six standard evaluation metrics.

◦

Curated and preprocessed **multi-audio** evaluation sets of **2-5** audio-image pairs from a 50-class VGG-Sound subset.

PROJECTS

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Adversarial Robustness of Vision Transformers
(Prof. Anirban Dasgupta, IIT Gandhinagar) | [Github](#) | [PyTorch](#), ViT

[Jan. '25 – Apr. '25]

◦

Evaluated adversarial robustness of **ViT** with sinusoidal, learnable, and RoPE positional embeddings.

◦

Proposed a **novel positional encoding** by concatenating positional and patch embeddings in a fixed ratio.

◦

Benchmarked over **400** ViT architectures on CIFAR-10 under black-box attack settings.

◦

Achieved **2% accuracy improvement** and outperformed standard positional encodings across attacks.

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Olympia – Sports Management System
(Prof. Yogesh Meena, IIT Gandhinagar) | [Github](#)

[Jan. '26 – Mar. '26]

◦

Created a web application using **FastAPI** and **MySQL** with **JWT** authentication, **RBAC**, and session management.

◦

Performed API endpoint benchmarking with **EXPLAIN analysis** and optimized query performance using **indexing**.

◦

Implemented **hash based** database **sharding** across multiple tables for horizontal scalability and load distribution.

TECHNICAL SKILLS

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Languages & Libraries: C, C++, C#, Python, NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch.

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Software & Tools: Git, GitHub, Xilinx Vivado, Autodesk Inventor, Arduino IDE, $\LaTeX$.

ACHIEVEMENTS

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Earned **Dean's List** honors in 1st and 2nd semesters (top 5%) and received an **Academic Citation** in the 3rd semester.

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Secured All-India JEE Advanced ranks of **1,544** in 2023 and **2,910** in 2022, placing within the top 2% nationwide.

POSITIONS OF RESPONSIBILITY

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Executive, Curation Department, TEDxIITGandhinagar

[Aug '24 – May '25]

◦

Built a database of **500+** potential speakers, drafted personalized outreach emails, and hosted faculty podcasts.

◦

Coordinated speaker logistics including travel, accommodation, and script development.

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Junior Sponsorship Executive, Amalthea, IIT Gandhinagar

[Aug '23 – Nov '23]

◦

Secured sponsorships through proposals, cold outreach, sponsor interaction, and post-event reports.